Stabilized Selection Algorithms:

Research working document

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Abstract

This document collects direct examples in which a data-dependent selection rule is stabilized, randomized, privatized, or otherwise certified. The aim is not to survey all of algorithmic stability, differential privacy, or selective inference. The aim is narrower: identify concrete algorithmic patterns of the form

$$\text{original selector} \to \text{stabilized selector} \to \text{formal certificate} \to \text{utility or inference cost.}$$

These examples serve as raw material for a possible composable stability-certification framework for post-hoc selection algorithms.

1 Introduction

Many post-hoc selection procedures choose a model, feature, winner, support set, hyperparameter, candidate solution, hypothesis, query, or arm based on the same data later used for evaluation or inference. This reuse induces dependence between the selected object and the inferential statistic. In simple settings this appears as the winner's curse; in model selection it appears as invalid naive confidence intervals; in adaptive validation it appears as overfitting to a holdout set.

Existing literatures contain many algorithm-specific fixes: randomized selective inference, differential privacy, noisy argmax, noisy top-k, sparse vector methods, stability selection, reusable holdout methods, and stability-based validation. This document does not claim that the literature already contains a universal composable framework. It identifies concrete examples that suggest such a framework may be possible.

Scope. A main example must have a concrete selection map, a selected object, a mechanism that stabilizes or certifies the selector, a formal guarantee, and an explicit cost. Foundational stability or privacy papers appear only as primitive references unless they instantiate a concrete selection primitive.

2 Inclusion Criteria and Audit

A paper is included as a **MAIN** example only if it supplies most of the following: a selector $\mathcal{A}(D)$, a selected object $\widehat{S}$, a stabilizing/certifying mechanism, a theorem-level guarantee, and a utility or inference cost. Papers that only give background stability theory, prediction generalization, lower bounds, or non-randomized comparison procedures are demoted to primitive reference, baseline, or excluded.

Table 1: Inclusion audit. Verification details are summarized separately in the companion notes.

Citation	Selected object	Selector	Mechanism	Guarantee
Zrnic and Jordan (2023) noisy winner	winner index	$\arg\max_i y_i$	Laplace noisy argmax	(η, τ, ν)-stability and CI correction
Zrnic and Jordan (2023) marginal screening	features	top-k $\lvert X_j^\top y \rvert$	per-step Laplace noise	stable selected model and CI correction
Zrnic and Jordan (2023) Stable LASSO	support/atoms	Frank–Wolfe atom path	Laplace noisy atom selection	composed stability
Tian and Taylor (2015)	active set/model	randomized response selection	added randomization	selective CLT/valid inference
Huang et al. (2024)	selected groups	group lasso	Gaussian objective randomization	selective likelihood/Wald regions
Markovic et al. (2017)	model/ hyperparameter	CV/AIC model choice	Gaussian randomized quality vectors	selective valid pivots
Meinshausen and Buehlmann (2010)	variables/ structure	base selector on subsamples	subsampling/ frequency threshold	PFER bound under assumptions
Shah and Samworth (2013)	variables	base selector on complementary pairs	paired subsampling	low/high selection-probability bounds
Bach (2008)	LASSO support	bootstrap LASSO supports	support intersection	model consistency
Faletto and Bien (2022)	clusters/features	stability selection under clusters	cluster aggregation	cluster-level analogues of SS/CPSS bounds
Dwork and Roth (2014) RNM	argmax index	max score	noisy scores	ϵ-DP
McSherry and Talwar (2007)	candidate item	utility maximizer	exponential mechanism	ϵ-DP utility selection
McKenna and Sheldon (2020)	candidate item	best score	permute-and-flip	DP and utility-loss dominance vs EM

Citation	Selected object	Selector	Mechanism	Guarantee
Qiao et al. (2021)	top-k set	top-k scores	one-shot Laplace	approximate DP for unordered top-k
Shekelyan and Loukides (2022)	top-k set	top-k scores	Lipschitz additive noise family	DP proof for unified family
Durfee and Rogers (2019)	top-k items	limited-domain top-k	Gumbel/threshold mechanisms	approximate DP, pay-what-you-get by returned outputs
Bun et al. (2018)	high-score item	gap-separated max	GAP-MAX	(ρ, ω)-tCDP stable selection under gap promise
Liu and Talwar (2019)	candidate/ model	select among private candidates	selector over ϵ_1-DP candidates	DP with controlled selection cost
Chaudhuri and Vinterbo (2013)	hyperparameter/ model	validation-set model choice	stability-based validation	private validation/tuning
Papernot and Steinke (2022)	hyperparameter	tuning over private runs	randomized tuning/accounting	RDP privacy accounting
Dwork et al. (2015) Thresholdout	adaptive query/ model statistic	holdout query test	noisy thresholding	adaptive generalization/holdout reuse
SVT/AT (Dwork and Roth, 2014)	threshold-passing query	first/limited passing queries	noisy threshold and query noise	DP
Blum and Hardt (2015), Hardt (2017)	leaderboard best-so-far	update if improvement clears margin	thresholded/noisy updates	reliable adaptive leaderboard
Rogers et al. (2016)	chosen hypothesis	adaptive hypothesis testing	max-information/DP correction	valid corrected p-values
Azize et al. (2024)	best arm	adaptive arm selection	private mean estimates/noise	private best-arm identification
Chaudhuri et al. (2011)	model parameter	ERM output	output/objective perturbation	DP ERM
Hardt et al. (2016)	hypothesis	SGD iterate	early stopping/small steps	uniform stability
Bousquet and Elisseeff (2002)	hypothesis	regularized ERM	strong regularization	uniform stability/generalization

Citation	Selected object	Selector	Mechanism	Guarantee
Lee et al. (2016)	LASSO active set	LASSO selection event	conditioning, no stabilization	exact selective inference
Berk et al. (2013)	selected model coefficient	arbitrary model selection	simultaneous coverage, no stabilization	PoSI valid intervals
Andrews et al. (2024), Zrnic and Fithian (2024)	winner	non-randomized argmax	conditional/zoom correction	winner inference

3 Unifying Template

Let

$$\widehat{S} = \mathcal{A}(D)$$

be a data-dependent selector. A stabilized selector has the form

$$\widetilde{S} = \widetilde{\mathcal{A}}(D; \xi),$$

where ξ is noise, subsampling randomness, a randomized optimization perturbation, a private mechanism, or another stabilizing device. A certificate may be an algorithmic-stability bound,

$$\mathrm{Cert}(\widetilde{\mathcal{A}}) \leq \eta,$$

a differential privacy guarantee,

$$\widetilde{\mathcal{A}} \text{ is } (\epsilon, \delta)\text{-DP},$$

or a post-selection inference statement,

$$\Pr\{\theta_{\widetilde{S}} \in C_{\widetilde{S}}(D)\} \geq 1 - \alpha.$$

The repeated pattern extracted from the examples is

$$\boxed{\text{primitive} + \text{mechanism} + \text{proof strategy} \Rightarrow \text{certificate at cost of utility/inference loss.}}$$

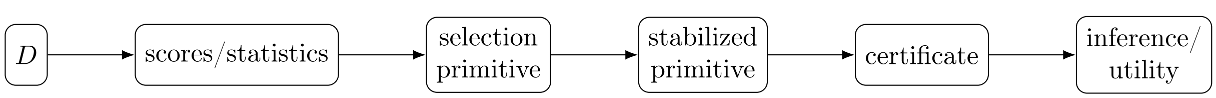

Figure 1: General template for a composable stability-certification module.

4 Master Summary Table

Table 2: Direct examples of stabilized or certified selection.

Paper	Original selector and object	Stabilized selector and mechanism	Certificate / guarantee	Cost and primitive
Zrnic and Jordan (2023)	winner: $\arg\max_i y_i$	noisy winner: $\arg\max_i(y_i + \xi_i)$; Laplace noise	stability $\to$ CI correction	selection noise; wider CIs; noisy argmax
Zrnic and Jordan (2023)	features: top-k $\lvert X_j^\top y \rvert$	noisy sequential top-k; Laplace noise	composed stability	top-k score loss; noisy top-k
Zrnic and Jordan (2023)	support: FW/LASSO path	noisy atom selection; Laplace noise	composed stability	excess risk; iterative atom selection
Tian and Taylor (2015)	active set: model selection	randomized response/optimization	selective CLT	selection rand./comp.; randomized SI
Huang et al. (2024)	groups: group lasso	randomized group objective; Gaussian noise	asymptotic selective likelihood	inference cost; group support
Markovic et al. (2017)	model: CV/AIC choice	randomized quality vectors; Gaussian noise	selective pivots	computation/width; validation comparison
Meinshausen and Buehlmann (2010)	variables: base selector	subsampling and frequency thresholding	PFER bound under exchangeability/random guessing	conservative set; aggregation
Shah and Samworth (2013)	variables: base selector	complementary-pair subsampling	low/high selection-probability bounds	repeated fits; aggregation
Bach (2008)	support: LASSO support	bootstrap supports, then intersection	consistency	possible false negatives; aggregation
Faletto and Bien (2022)	clusters/features: stability selection	cluster-level aggregation; subsampling	cluster-level analogues of SS/CPSS bounds	chosen partition; gains depend on cluster quality
Dwork and Roth (2014)	index: maximum score	report noisy max; sensitivity-calibrated Laplace noise	DP	rank error; noisy argmax

Paper	Original selector and object	Stabilized selector and mechanism	Certificate / guarantee	Cost and primitive
McSherry and Talwar (2007)	candidate: utility maximizer	exponential mechanism; exponential sampling	DP	utility regret; discrete selection
McKenna and Sheldon (2020)	item: best score	permute-and-flip; randomized scan	DP	utility-loss stochastic dominance vs EM; discrete selection
Qiao et al. (2021)	top-k set: top-k scores	one-shot noisy top-k; Laplace noise	approximate DP	theorem-scale Laplace noise; top-k
Shekelyan and Loukides (2022)	top-k set: top-k scores	Lipschitz additive noise family	DP	noise/opt. cost; top-k family
Durfee and Rogers (2019)	items: unknown-domain top-k	limited-domain thresholded top-k; Gumbel/threshold	approximate DP with pay-what-you-get composition	may return fewer items; top-k
Bun et al. (2018)	item: high-score item	gap-aware selection; noise plus gap test	(ρ, ω)-tCDP stable selection under gap promise	requires score gap; gap argmax
Liu and Talwar (2019)	candidate/model: best candidate	select among ϵ_1-DP candidate mechanisms with bounded scores	DP	candidate cost; candidate selection
Chaudhuri and Vinterbo (2013)	hyperparameter: validation winner	stability-based validation; private/stable scoring	DP validation	budget cost; validation choice
Papernot and Steinke (2022)	hyperparameter: best private run	non-adaptive random search; random repetitions	RDP accounting	trial/privacy cost; tuning
Dwork et al. (2015)	query/statistic: adaptive holdout query	Thresholdout; noisy threshold	max-info/gen.	coarse feedback; reusable holdout
Dwork and Roth (2014)	query: threshold crossing	Sparse Vector; noisy threshold	DP	pass/fail noise; threshold primitive
Blum and Hardt (2015), Hardt (2017)	submission: best leaderboard update	threshold/noisy update; margin/noise	reliable leaderboard	reduced feedback; validation update

Paper	Original selector and object	Stabilized selector and mechanism	Certificate / guarantee	Cost and primitive
Rogers et al. (2016)	hypothesis: adaptively chosen test	max-information correction; DP/max-info	valid p-values	conservative tests; inference bridge
Azize et al. (2024)	arm: best arm	private mean estimators; private stopping/analysis	private BAI	sample complexity; best arm

5 Direct Examples

Each subsection below follows the same template: original selector, problem, stabilized version, mechanism, guarantee, proof idea, cost, and extracted primitive.

5.1 Algorithmic-stability post-selection inference: noisy winner

Citation. Zrnic and Jordan (2023), key `zrnic2023postselection`. Verified against the uploaded PDF.

Original selection algorithm. The canonical winner selector is

$$\widehat{i} = \arg\max_{i \in [m]} y_i.$$

Problem. Inference on the selected coordinate is biased by the winner's curse because the same data both select and estimate the target.

Stabilized version. A randomized selector adds independent Laplace noise:

$$\widetilde{i} = \arg\max_{i} \{y_i + \xi_i\}, \qquad \xi_i \sim \mathrm{Lap}(b).$$

The paper defines indistinguishability by $Q \approx_{\eta,\tau} W$ when $\Pr(Q \in O) \le e^{\eta} \Pr(W \in O) + \tau$ for all measurable O, and defines (η, τ, ν)-stability by comparison to an oracle selection with high probability (Zrnic and Jordan, 2023).

Formal guarantee. If $\widehat{S}$ is (η, τ, ν)-stable, then fixed-target intervals with nominal miscoverage $\delta e^{-\eta}$ satisfy

$$\Pr\{\theta_{\widehat{S}} \notin C_{\widehat{S}}(\delta e^{-\eta})\} \le \delta + \tau + \nu$$

(Zrnic and Jordan, 2023, Theorem 2).

Proof idea. Laplace density ratios satisfy a shift bound of the form $f_\xi(z - x)/f_\xi(z) \le e^{|x|/b}$. Calibrating b to the concentration of $y - \mu$ makes the randomized selector close in max-divergence to an oracle selector independent of y.

Cost. Smaller η gives less CI inflation but requires more noise and reduces probability of selecting the empirical winner.

noisy argmax + Laplace density-ratio proof $\Rightarrow$ (η, τ, ν)-stable selection and valid post-selection CIs at the cost of selection noise and interval inflation.

Framework relevance: high.

5.2 Algorithmic-stability post-selection inference: marginal screening

Citation. Zrnic and Jordan (2023), key `zrnic2023postselection`. Verified against the uploaded PDF.

Original selection algorithm. Marginal screening selects the k features with largest correlations:

$$\widehat{M} = \mathrm{TopK}_j\{|X_j^\top y| : j \in [d]\}.$$

Problem. Selecting the largest correlations and then constructing intervals for those selected coordinates reuses y.

Stabilized version. The stable marginal-screening algorithm computes normalized correlations c_j and repeatedly selects

$$i_t = \arg\max_{i \in R_t}\{|c_i| + \xi_{t,i}\}, \qquad \xi_{t,i} \sim \mathrm{Lap}(b),$$

removing selected coordinates after each step.

Formal guarantee. Algorithm 3 has both a refined composed-stability guarantee,

$$\left(\frac{1}{2}k\eta^2 + \sqrt{2k\log(1/\delta)}\,\eta, \delta, \delta\right)\text{-stability},$$

and the simpler $(k\eta, 0, \delta)$-stability guarantee. The same displayed δ is the chosen slack level used in the proposition-level summary; the three coordinates are the composed stability parameter, the indistinguishability slack, and the oracle-failure probability. Proposition 5 gives the corresponding top-k utility bound controlling how far selected noisy correlations fall below the true top-k correlations (Zrnic and Jordan, 2023, Propositions 4–5).

Proof idea. The proof parallels noisy max plus adaptive composition: each step is a noisy argmax over remaining correlations, and post-processing gives stability of the returned model.

Cost. The utility loss is a noisy top-k score-gap bound, controlled uniformly over the selected ranks with scale proportional to the Laplace noise and logarithmic factors such as $\log(dk/\delta')$; the stability parameter composes over the k adaptive selections.

sequential noisy top-k + Laplace report-noisy-max proof + adaptive composition $\Rightarrow$ stable feature screening at the cost of rank/score loss.

Framework relevance: high.

5.3 Algorithmic-stability post-selection inference: Stable LASSO

Citation. Zrnic and Jordan (2023), key `zrnic2023postselection`. Verified against the uploaded PDF.

Original selection algorithm. The target is LASSO-style support selection, approximated by a Frank–Wolfe path over atoms $\phi \in C_1\{\pm e_i\}$.

Problem. The support of a LASSO fit is discontinuous in the data and is used for downstream inference.

Stabilized version. Algorithm 2 in the paper samples Laplace noise for each atom at each step and chooses the noisy best atom, then performs a Frank–Wolfe update:

$$\phi_t = \operatorname*{arg\,min}_{\phi \in C_1\{\pm e_i\}} \{\alpha_\phi(y, \theta_t) + \xi_{t,\phi}\}.$$

Here $\alpha_\phi(y, \theta_t)$ denotes the deterministic Frank–Wolfe atom score at iterate θ_t, and $\xi_{t,\phi}$ is the independent Laplace perturbation for atom ϕ at step t.

Formal guarantee. Algorithm 2 has both a refined composed-stability guarantee,

$$\left(\frac{1}{2}k\eta^2 + \sqrt{2k\log(1/\delta)}\,\eta, \delta, \delta\right)\text{-stability},$$

and the simpler $(k\eta, 0, \delta)$-stability guarantee. As above, the displayed δ is the chosen slack level in this compact proposition-level statement. Stability transfers to the selected support by post-processing (Zrnic and Jordan, 2023, Proposition 2).

Proof idea. Each noisy atom-selection step is proved stable by an extension of Report Noisy Max, then composed over k adaptive steps. The selected support is a post-processing of the stable iterate.

Cost. Randomization introduces an excess-risk term relative to the constrained LASSO solution; Proposition 3 gives an explicit constrained least-squares excess-risk bound depending on C_1, $\|X\|_{2,\infty}$, logarithmic factors in d, the noise/concentration scale, n, and η.

$$\boxed{\text{iterative atom selection + per-step Laplace noisy argmax + adaptive composition} \Rightarrow \text{stable support selection at cost of excess optimization risk.}}$$

Framework relevance: high.

5.4 Randomized LASSO selective inference

Citation. Tian and Taylor (2015), key `tian2015randomized`. Verified against theorem-level statements in the primary source.

Original selection algorithm. The original selector is LASSO or another model-selection procedure returning an active set.

Problem. Conditioning on a non-randomized selection event can be overly restrictive; ignoring the event is invalid.

Stabilized version. The procedure injects external randomization into the response or selection statistic before model selection. Abstractly,

$$\widetilde{M} = \mathcal{A}(y + \omega), \qquad \omega \sim Q,$$

or equivalently a randomized optimization problem is solved.

Formal guarantee. Under independent sampling, fixed-dimensional or linearizable-statistic assumptions, smoothness and tail conditions, and lower bounds on the randomized selection probability, the paper proves selective or privatized CLTs yielding asymptotically uniform pivots for randomized selective inference (Tian and Taylor, 2015).

Proof idea. Randomization smooths the selection event and allows selective likelihood and weak-convergence arguments.

Cost. Randomization is used in selection, while inference is carried out on the original data conditional on the randomized selection event. The cost is possible degradation of selection quality and additional computation; randomization can also improve power relative to non-randomized conditioning by leaving more Fisher information for inference.

support/model selector + external randomization + selective likelihood/CLT $\Rightarrow$ asymptotically valid selective inference at cost of selection randomization and computation.

Framework relevance: high.

5.5 Randomized group LASSO selective inference

Citation. Huang et al. (2024), key `huang2024randomizedgroup`. Verified against theorem-level statements in the primary source.

Original selection algorithm. Group LASSO selects groups of covariates through group-regularized optimization.

Problem. Post-selection inference for selected grouped covariates is biased if group selection is ignored.

Stabilized version. A randomized group-regularized optimization problem is studied. Suppressing model-specific notation but retaining the asymptotic scaling, the randomized objective has the form

$$\widetilde{\beta} = \arg\min_{\beta} \left\{ n^{-1/2}\ell_D(\beta) + \sum_{g \in G} \lambda_g \|\beta_g\|_2 - \sqrt{n}\,\omega_n^\top \beta \right\}, \qquad \sqrt{n}\,\omega_n \sim N_p(0, \Omega).$$

Here G is the fixed group collection, β_g is the coefficient block for group g, λ_g are group penalties, ℓ_D is the empirical loss or negative log-likelihood, and Ω is the Gaussian randomization covariance.

Formal guarantee. Under the paper's asymptotic smoothness, density, moment, and large-deviation assumptions, and conditioning on selection-related quantities such as group directions and subgradients, the method constructs an asymptotic post-selection likelihood for the selected groups and Wald-type selective confidence regions with bounded inverse-information entries (Huang et al., 2024).

Proof idea. Gaussian randomization makes the randomized optimization variables amenable to a selective likelihood conditioning argument after accounting for the group-lasso KKT map.

Cost. Costs are randomization, model-specific assumptions, and optimization/inference computation; region width is a theorem-dependent inference cost rather than an unconditional monotone penalty.

group support selection + Gaussian objective randomization + selective likelihood ⇒ asymptotically valid selected-group inference under stated assumptions.

Framework relevance: high.

5.6 Randomized model-quality selection for cross-validation and AIC

Citation. Markovic et al. (2017), key `markovic2017crossvalidation`. Verified against theorem-level statements in the primary source.

Original selection algorithm. A model is chosen by minimizing cross-validation error, AIC, prediction error, or another quality vector:

$$\widehat{M} = \arg\min_{M \in \mathcal{M}} \widehat{R}_M.$$

Problem. Inference for a parameter chosen after model selection is invalid without accounting for the selection of $\widehat{M}$.

Stabilized version. The paper proposes adding Gaussian randomization to quality vectors or test statistics so the joint vector used for selection and inference is asymptotically multivariate Gaussian.

Formal guarantee. Under joint asymptotic Gaussian assumptions, affine selection representations, and fixed/low-dimensional regimes considered in the paper, it derives pivotal quantities for selectively valid tests and confidence intervals after procedures such as cross-validated LASSO and AIC-based selection (Markovic et al., 2017).

Proof idea. Convert the selection/inference statistics to a Gaussian selection model, then condition on the randomized selection event.

Cost. Computation, including MCMC as a device for some randomized model-selection intervals, and possible interval widening.

validation/model-quality comparison + Gaussian randomization + Gaussian selective pivot ⇒ selectively valid inference at cost of computation and wider intervals.

Framework relevance: high.

5.7 Stability selection

Citation. Meinshausen and Buehlmann (2010), key `meinshausen2010stability`. Verified against theorem-level statements in the primary source.

Original selection algorithm. A base selector $\mathcal{A}_\lambda(D)$ returns a set of variables.

Problem. High-dimensional feature selection is unstable; small sample perturbations can change the selected support.

Stabilized version. Draw half-subsamples $D^{(b)}$, run the base selector, and estimate selection frequencies. In the single-λ simplification,

$$\hat{\pi}_j = \frac{1}{B}\sum_{b=1}^{B} \mathbf{1}\{j \in \mathcal{A}_\lambda(D^{(b)})\}, \qquad \widehat{S} = \{j : \hat{\pi}_j \geq \pi_{\text{thr}}\}.$$

Formal guarantee. Under exchangeability of noise-variable selections and the "not worse than random guessing" condition, stability selection controls the expected number of false selections (PFER), not ordinary FDR. If V is the number of false selected variables among p candidates and q_Λ is the expected number of variables selected by the base procedure over the tuning grid Λ, then

$$\mathbb{E}[V] \leq \frac{q_\Lambda^2}{(2\pi_{\text{thr}} - 1)p}, \qquad \pi_{\text{thr}} > 1/2.$$

The original formulation maximizes selection probabilities over $\lambda \in \Lambda$; FWER follows only by a further Markov bound (Meinshausen and Buehlmann, 2010).

Proof idea. Aggregate repeated randomized/subsampled selections and bound false selections by selection probabilities.

Cost. More computation and conservative selection; lower false positives may come with lower power.

subsample aggregation + frequency threshold + selection-probability bounds ⇒ stable feature selection at cost of conservativeness.

Framework relevance: high.

5.8 Complementary Pairs Stability Selection

Citation. Shah and Samworth (2013), key `shah2013cpss`. Verified against theorem-level statements in the primary source.

Original selection algorithm. CPSS begins with any base variable selector.

Problem. Original stability selection bounds depend on assumptions such as exchangeability or can be conservative.

Stabilized version. Use complementary half-sample pairs and aggregate selections across pairs.

Formal guarantee. CPSS controls the expected number of selected variables with low base-procedure selection probability and, dually, the expected number of excluded variables with high selection probability. The worst-case bounds require no exchangeability or random-guessing assumptions on the base procedure; sharper bounds require shape restrictions such as unimodality or r-concavity (Shah and Samworth, 2013).

Proof idea. Paired subsamples create dependence structure that enables finite-sample expectation bounds without strong modeling assumptions.

Cost. Repeated fitting and conservative thresholds.

paired subsampling + base selector + frequency aggregation $\Rightarrow$ finite-sample error control at cost of repeated computation.

Framework relevance: high.

5.9 Bolasso

Citation. Bach (2008), key `bach2008bolasso`. Verified against theorem-level statements in the primary source.

Original selection algorithm. LASSO support selection returns $\widehat{S}(D) = \mathrm{supp}(\widehat{\beta}_\lambda(D))$.

Problem. With certain regularization rates, LASSO may include noise variables with positive probability even while selecting true variables with high probability.

Stabilized version. Run LASSO over bootstrap replications and intersect supports:

$$\widehat{S}_{\mathrm{Bolasso}} = \bigcap_{b=1}^{B} \mathrm{supp}(\widehat{\beta}_\lambda(D^{*(b)})).$$

Formal guarantee. The paper proves model consistency under fixed-p assumptions (A1–A3), $\mu_n = \mu_0 n^{-1/2}$ with $\mu_0 > 0$, and bootstrap count B satisfying $\log B \to \infty$ and $\log B = o(n)$; Proposition 3 gives an explicit upper bound on $\Pr(\widehat{S}_{\mathrm{Bolasso}} \neq J)$ (Bach, 2008).

Proof idea. True variables are selected with probability tending to one across bootstraps, whereas noise variables are removed by intersection.

Cost. Intersection can be conservative and computationally expensive.

bootstrap supports + intersection + asymptotic support-probability separation $\Rightarrow$ consistent model selection at cost of possible underselection.

Framework relevance: high.

5.10 Cluster Stability Selection

Citation. Faletto and Bien (2022), key `faletto2022cluster`. Verified against theorem-level statements in the primary source.

Original selection algorithm. Ordinary stability selection selects individual features by subsample frequency.

Problem. With highly correlated proxy features, LASSO may select only one proxy per subsample; frequency can be split across proxies so ordinary stability selection selects none.

Stabilized version. Use cluster-level aggregation, optionally combining features in selected clusters with weights derived from selection frequencies.

Formal guarantee. The paper gives cluster-level analogues of stability-selection and CPSS error-control bounds for a fixed partition; the bounds hold for any partition, while practical gains depend on whether the partition captures meaningful correlated-feature structure (Faletto and Bien, 2022).

Proof idea. Move the aggregation unit from individual features to clusters to avoid vote splitting, then apply stability-selection style bounds at the cluster level.

Cost. The formal guarantees depend on the chosen partition; ranking and prediction gains depend on cluster quality and the correlated-feature structure.

$$\boxed{\text{clustered support primitive + subsample frequency aggregation} \Rightarrow \text{cluster-level error control for a chosen partition, with practical gains depending on cluster quality.}}$$

Framework relevance: high.

5.11 Report Noisy Max

Citation. Dwork and Roth (2014), key `dwork2014book`. Verification from standard DP monograph/source dataset.

Original selection algorithm. Select the highest score:

$$\widehat{i} = \arg\max_i s_i(D).$$

Problem. The identity of the maximum can be sensitive to one individual or one data perturbation.

Stabilized version. Add noise to each score and return the noisy maximizer:

$$\widetilde{i} = \arg\max_i \{s_i(D) + \xi_i\}.$$

Formal guarantee. For a fixed candidate set and data-independent tie-breaking, releasing only the noisy maximizer after adding independent noise calibrated to score sensitivity is differentially private. For arbitrary sensitivity-Δ scores, independent $\mathrm{Lap}(2\Delta/\epsilon)$ noise is a standard ϵ-DP calibration; smaller constants can apply in monotone/count-query variants (Dwork and Roth, 2014).

Proof idea. The sensitivity/noise calibration bounds likelihood ratios of the selected index under neighboring datasets.

Cost. The selected item may fail to be the exact empirical maximizer.

argmax + score noise + sensitivity proof $\Rightarrow$ private/stable winner selection at cost of selection regret.

Framework relevance: high.

5.12 Exponential Mechanism

Citation. McSherry and Talwar (2007), key `mcsherry2007exponential`.

Original selection algorithm. Choose the candidate maximizing a data-dependent utility $u(D, r)$.

Problem. Exact utility maximization can leak information or be unstable.

Stabilized version. Sample r with probability proportional to

$$\exp\left(\frac{\epsilon u(D, r)}{2\Delta u}\right).$$

Formal guarantee. For a fixed, data-independent finite range $\mathcal{R}$ and bounded utility sensitivity Δu, the mechanism is ϵ-differentially private and gives utility guarantees relative to the best candidate. In the finite-range bound, with probability at least $1 - \beta$, the utility loss is at most $(2\Delta u/\epsilon)(\log|\mathcal{R}| + \log(1/\beta))$ up to the standard convention for the utility optimum (McSherry and Talwar, 2007).

Proof idea. Bound the change in the exponential weights under neighboring datasets using utility sensitivity.

Cost. Utility regret relative to exact maximization.

discrete candidate selection + exponential sampling + utility-sensitivity proof $\Rightarrow$ DP selection at cost of utility regret.

Framework relevance: high.

5.13 Permute-and-Flip

Citation. McKenna and Sheldon (2020), key `mckenna2020permute`. Verified against theorem-level statements in the primary source.

Original selection algorithm. Select a candidate with the highest quality score.

Problem. Private selection should preserve privacy while retaining high-score utility.

Stabilized version. Permute-and-Flip randomly orders candidates and accepts candidate r with probability

$$\exp\left(\frac{\epsilon(q_r - q^*)}{2\Delta}\right),$$

where q^* is the best score and Δ is the score sensitivity.

Formal guarantee. The paper proves that Permute-and-Flip is ϵ-DP for finite candidate sets with score sensitivity at most Δ, and that its utility-loss random variable is stochastically no larger than that of the exponential mechanism under the same score vector and privacy parameter; in particular, the expected score loss can improve by up to a factor of two (McKenna and Sheldon, 2020).

Proof idea. Analyze the privacy constraints directly and construct a mechanism satisfying them with improved utility.

Cost. Still pays privacy-induced selection regret, though less than the exponential mechanism in the paper's comparison.

$$\boxed{\text{finite candidate selection + permuted randomized accept/reject + privacy-constraint proof} \Rightarrow \text{DP selection with improved expected utility.}}$$

Framework relevance: high.

5.14 Oneshot differentially private top-k

Citation. Qiao et al. (2021), key `qiao2021oneshot`. Verified against theorem-level statements in the primary source.

Original selection algorithm. Return the top-k elements by score.

Problem. Running Report Noisy Max k times pays a sequential-composition cost.

Stabilized version. Add Laplace noise once to all scores and compute the top-k elements once.

Formal guarantee. For score sensitivity s_f, $\epsilon \leq 0.2$, $\delta \leq 0.05$, and $m \geq 2$, the one-shot Laplace mechanism is (ϵ, δ)-DP when

$$\lambda \geq \frac{8 s_f \sqrt{k \log(m/\delta)}}{\epsilon}.$$

The same paper proves pure DP with $\lambda \geq 2k s_f/\epsilon$. The approximate-DP result is obtained by a coupling argument rather than by composing k peeling/noisy-max calls (Qiao et al., 2021).

Proof idea. A novel coupling argument controls the top-k selected set directly rather than composing k noisy max calls.

Cost. Noise grows with $\sqrt{k}$ and can reduce top-k recall.

$$\boxed{\text{top-}k\text{ selection + one-shot Laplace perturbation + coupling proof} \Rightarrow \text{approximate DP without } k\text{-fold peeling cost.}}$$

Framework relevance: high.

5.15 Canonical Lipschitz Mechanism

Citation. Shekelyan and Loukides (2022), key `shekelyan2022lipschitz`. Verified against theorem-level statements in the primary source.

Original selection algorithm. Select the top-k scoring items from d items.

Problem. Many private top-k mechanisms are analyzed separately and sequential composition can hurt utility.

Stabilized version. Use an additive-noise family whose survival function satisfies the log-Lipschitz condition that $\log(1 - F(x))$ is 1-Lipschitz, scaled by the relevant sensitivity/privacy parameters; pair it with a canonical loss over finite selection domains such as subsets.

Formal guarantee. Under that log-survival Lipschitz condition and the finite-sensitivity loss setup, the paper unifies exponential mechanism, permute-and-flip, Report Noisy Max, and oneshot variants under one DP proof, and gives a composition-free top-k subset selection approach (Shekelyan and Loukides, 2022).

Proof idea. A Lipschitz property of the noise survival function yields a uniform privacy argument across mechanisms.

Cost. Computational and noise choices determine top-k utility.

top-k subset selection + Lipschitz additive noise family + unified DP proof $\Rightarrow$ reusable private selection module.

Framework relevance: high.

5.16 Practical DP top-k with pay-what-you-get composition

Citation. Durfee and Rogers (2019), key `durfee2019topk`. Verified against theorem-level statements in the primary source.

Original selection algorithm. Select top-k elements from a large or unknown domain.

Problem. Naive private top-k can require querying the whole domain and paying for every possible output.

Stabilized version. Use limited-domain access and Gumbel/threshold-style selection, with algorithms that may return fewer than k elements.

Formal guarantee. The paper gives approximate-DP limited-domain top-k algorithms and a pay-what-you-get composition theorem: across at most ℓ^* adaptive top-k queries, with total returned-output budget k^*, Algorithm 2 is $(\epsilon^*, 2\ell^*\delta + \delta_0)$-DP, where

$$\epsilon^* = \min\left\{k^*\epsilon,\ k^*\epsilon\frac{e^\epsilon - 1}{e^\epsilon + 1} + \epsilon\sqrt{2k^*\log(1/\delta_0)},\ \frac{k^*\epsilon^2}{2} + \epsilon\sqrt{\frac{k^*\log(1/\delta_0)}{2}}\right\},$$

with the two $\log(1/\delta_0)$ terms used for $\delta_0 > 0$ and the basic-composition term always available (Durfee and Rogers, 2019, Theorem 2).

Proof idea. Relate the selection to exponential-mechanism/noisy-threshold ideas and account adaptively for actual returned outputs.

Cost. May return fewer than k elements and requires threshold tuning.

unknown-domain top-k + thresholded/Gumbel selection + output-size accounting $\Rightarrow$ practical DP top-k at cost of possible misses.

Framework relevance: high.

5.17 GAP-MAX

Citation. Bun et al. (2018), key `bun2018gapmax`. Verified against theorem-level statements in the primary source.

Original selection algorithm. Select an item with the largest low-sensitivity score.

Problem. Worst-case private selection can be overly costly when many candidates exist.

Stabilized version. GAP-MAX/stable selection considers sensitivity-1 scores $q_1, \ldots, q_k$ and promises a unique maximizer i^* with

$$q_{i^*}(x) \geq \max_{i \neq i^*} q_i(x) + \mathrm{GAP}.$$

Formal guarantee. Privacy comes from the chosen noisy or thresholded mechanism and from composition/post-processing. For $\rho \in (0, 1)$ and $\omega \geq 1/\sqrt{\rho}$, the paper proves, up to universal constants, a (ρ, ω)-tCDP stable-selection algorithm when

$$\mathrm{GAP} \geq O\left(\omega \log\left(1 + \frac{\sqrt{\log k}}{\omega\sqrt{\rho}}\right)\right),$$

The same source compares corresponding gap requirements for pure-DP, zCDP, and approximate-DP regimes (Bun et al., 2018).

Proof idea. Use a privacy-preserving noisy/thresholded selection mechanism, then analyze how a score gap improves the probability of returning the true maximizer.

Cost. Requires a sufficiently large gap; when candidates are close, no large improvement is available.

argmax + private noisy/thresholded selection + gap-dependent accuracy analysis $\Rightarrow$ private winner selection that can avoid worst-case utility cost under a gap promise.

Framework relevance: high.

5.18 Private Selection from Private Candidates

Citation. Liu and Talwar (2019), key `liu2019privatecandidates`. Verified against theorem-level statements in the primary source.

Original selection algorithm. Select the best candidate or hyperparameter after generating candidate models/procedures.

Problem. Selecting among many candidates can consume privacy/stability budget if every candidate is treated independently.

Stabilized version. Candidate generators or scores are already private; the paper studies mechanisms for selecting among private candidates with controlled additional privacy cost.

Formal guarantee. The candidates are randomized mechanisms $M_i(D)$ that are already ϵ_1-DP, and each oracle call returns a sample together with a bounded score, typically in $[0, 1]$. The paper gives selectors with privacy costs $2\epsilon_1$, $3\epsilon_1$, or $(2\epsilon_1 + \epsilon_0, \delta)$ depending on whether a threshold is known, random stopping is used, or the more elaborate near-$2\epsilon_1$ selector is used (Liu and Talwar, 2019).

Proof idea. The selector exploits the fact that candidate-producing or scoring procedures are already DP randomized mechanisms with bounded scores, then uses thresholding, competition, and randomization to choose among private candidates without paying a full composition cost for every candidate.

Cost. Requires candidates to be generated privately and incurs selection overhead.

$$\boxed{\begin{array}{c}\text{private candidates + private selector + composition accounting} \Rightarrow \text{stable/private} \\ \text{hyperparameter or model choice at cost of candidate-generation budget.}\end{array}}$$

Framework relevance: high.

5.19 Stability-based validation for differentially private machine learning

Citation. Chaudhuri and Vinterbo (2013), key `chaudhuri2013validation`. Verified against theorem-level statements in the primary source.

Original selection algorithm. Choose a hyperparameter or model by validation score.

Problem. Hyperparameter selection over many candidates can spend privacy budget and overfit validation data.

Stabilized version. Use private training together with a validation procedure whose validation score satisfies explicit stability assumptions over the searched parameter list.

Formal guarantee. Under the paper's $(\beta_1, \beta_2, \delta/k)$-style stability assumptions for the validation scores over k parameter values, VALIDATE is (α_2, δ)-DP; retraining with the selected parameter gives $(\alpha_1 + \alpha_2, \delta)$-DP if the training algorithm is α_1-DP (Chaudhuri and Vinterbo, 2013).

Proof idea. Stability of the validation score and privacy of the training algorithm permit controlled selection among a finite list of parameter settings; this is not a generic guarantee for arbitrary validation metrics or arbitrary private training pipelines.

Cost. Privacy budget and validation accuracy trade off with the breadth of the hyperparameter search.

validation-set comparison + stable validation score + private training $\Rightarrow$ private hyperparameter selection at cost of validation/privacy budget.

Framework relevance: high.

5.20 Rényi-DP hyperparameter tuning

Citation. Papernot and Steinke (2022), key `papernot2022rdp`. Verified against theorem-level statements in the primary source.

Original selection algorithm. Select hyperparameters after repeated private training/evaluation runs.

Problem. Hyperparameter tuning can itself leak information and consume privacy budget.

Stabilized version. Use non-adaptive random search over a finite candidate set with a random number of repetitions and uniform RDP bounds for the base private mechanism.

Formal guarantee. The source develops Rényi-DP accounting for hyperparameter tuning in this randomized random-search setting; deterministic exhaustive grid search or adaptive tuning is not covered by the improved random-repetition theorem except through ordinary composition or separate analysis (Papernot and Steinke, 2022).

Proof idea. Randomizing the number of private training/evaluation runs changes the privacy accounting for selecting the best run from a totally ordered output space.

Cost. More trials spend privacy budget under the applicable RDP accounting; deterministic or adaptive tuning can fall back to worse ordinary-composition costs.

non-adaptive random search + random repetitions + RDP accounting $\Rightarrow$ privacy-certified tuning at cost of trial/privacy budget.

Framework relevance: high for the non-adaptive random-search setting.

5.21 Thresholdout and reusable holdout

Citation. Dwork et al. (2015), key `dwork2015adaptive`. Verified against theorem-level statements in the primary source.

Original selection algorithm. Adaptive analysts repeatedly query a validation/holdout set and use the answers to choose models or statistics.

Problem. Repeated adaptive validation overfits the holdout.

Stabilized version. Thresholdout returns the training-set estimate unless a noisy discrepancy test detects overfitting between training and holdout estimates; when the noisy threshold is crossed, it returns a noisy holdout estimate and consumes holdout budget.

Formal guarantee. The work connects differential privacy, approximate max-information, and generalization to enable adaptive holdout reuse (Dwork et al., 2015).

Proof idea. Privacy/generalization transfer: algorithms with bounded information leakage about the sample generalize even under adaptive queries.

Cost. Coarse/noisy feedback, conservative thresholds, and holdout budget consumption.

adaptive validation query + noisy thresholding + max-information/DP transfer $\Rightarrow$ reusable holdout at cost of coarse feedback.

Framework relevance: high.

5.22 Sparse Vector and AboveThreshold

Citation. Dwork and Roth (2014), key `dwork2014book`.

Original selection algorithm. Scan adaptive low-sensitivity queries and report those exceeding a threshold.

Problem. Reporting threshold crossings can leak information about many query answers.

Stabilized version. Add noise to the threshold and/or query answers and release limited passing events.

Formal guarantee. Sparse Vector/AboveThreshold is a DP threshold-selection primitive under sensitivity-calibrated noise (Dwork and Roth, 2014).

Proof idea. The privacy proof uses a noisy threshold and noisy query comparisons, with privacy cost controlled by a preset cap on positive releases; accuracy depends on margins to the threshold, so near-threshold queries can still affect error behavior.

Cost. False positives/negatives around the threshold and limited number of positive releases.

threshold selection + noisy threshold/query values + sparse-vector proof $\Rightarrow$ adaptive DP thresholding at cost of pass/fail errors.

Framework relevance: high.

5.23 Ladder and Shaky Ladder

Citation. Blum and Hardt (2015) and Hardt (2017). Verified against theorem-level statements in the primary sources.

Original selection algorithm. Public leaderboards repeatedly reveal validation scores, allowing adaptive overfitting.

Problem. Participants adapt submissions to leaderboard feedback, causing validation leakage.

Stabilized version. Ladder reveals a new leaderboard score only when a submission improves over the current best by a threshold; Shaky Ladder is a randomized Ladder variant for adaptive risk estimation.

Formal guarantee. Ladder provides leaderboard-accuracy guarantees for best-so-far submissions; Shaky Ladder improves the adaptive leaderboard-error rate under its assumptions, not risk accuracy for every submitted model.

Proof idea. Restrict information release through thresholded/noisy updates, limiting adaptive overfitting.

Cost. Less feedback and coarser scores.

leaderboard update + threshold/noisy improvement gate $\Rightarrow$ stable adaptive validation at cost of reduced feedback.

Framework relevance: high.

5.24 Max-information and post-selection hypothesis testing

Citation. Rogers et al. (2016), key `rogers2016maxinformation`. Verified against theorem-level statements in the primary source.

Original selection algorithm. An adaptive analysis chooses a hypothesis/test based on data-dependent outputs.

Problem. Naive p-values after adaptive hypothesis choice are invalid.

Stabilized/certified version. Use bounded approximate max-information, often obtained from differential privacy, to correct p-values for adaptively chosen hypotheses.

Formal guarantee. The paper shows that bounded approximate max-information suffices for post-selection p-value corrections, and that approximate DP implies bounded approximate max-information under product input distributions. The DP-to-testing implication therefore requires the max-information bridge and is sensitive to composition order (Rogers et al., 2016).

Proof idea. Transfer privacy/generalization stability into control of adaptive testing error.

Cost. More conservative corrected p-values; composition subtleties for approximate DP.

adaptive hypothesis selection + max-information certificate $\Rightarrow$ valid post-selection testing at cost of conservative p-value correction.

Framework relevance: high as a bridge, not a selector mechanism by itself.

5.25 Differentially private best-arm identification

Citation. Azize et al. (2024), key `azize2024dpbai`. Verified against theorem-level statements in the primary source.

Original selection algorithm. A bandit algorithm adaptively samples and selects the best arm.

Problem. Arm means and the identity of the best arm can leak individual reward information.

Stabilized version. DP best-arm algorithms privatize mean estimates and use private stopping/selection analyses before identifying the best arm.

Formal guarantee. The paper studies fixed-confidence, δ-correct best-arm identification under local and global/central ϵ-DP, proving lower bounds and sample-complexity tradeoffs for private Top-Two style algorithms built from private mean-estimation primitives (Azize et al., 2024).

Proof idea. Combine private mean estimators with adaptive sampling, stopping, and change-of-measure analyses for best-arm identification.

Cost. Increased sample complexity and privacy-budget cost for a fixed target error probability.

$$\boxed{\text{best-arm selection + private mean-estimation primitives + adaptive stopping/selection analysis} \Rightarrow \text{private arm identification at cost of sample complexity.}}$$

Framework relevance: medium-high.

6 Primitive References

Laplace and Gaussian mechanisms. The sensitivity-calibrated noise paradigm from Dwork et al. (2006), and the Laplace/Gaussian mechanisms as summarized by Dwork and Roth (2014), are not themselves selection algorithms, but they supply proof templates for noisy scalar release, noisy argmax, thresholding, and several private selection mechanisms.

Objective and output perturbation. Private ERM via output and objective perturbation (Chaudhuri et al., 2011) stabilizes an optimization solve. It is not necessarily a post-hoc selector, but it becomes relevant when the selected object is a model or support derived from a private optimizer.

Regularized ERM stability. Classical uniform stability for regularized learning (Bousquet and Elisseeff, 2002) and trajectory stability of SGD (Hardt et al., 2016) are not main examples here because they generally stabilize prediction/generalization rather than selection. They provide proof templates for optimization primitives in a future framework.

Composition and post-processing. DP composition and post-processing theorems, as summarized by Dwork and Roth (2014), are essential for composing private primitives and for preserving DP under functions of private outputs; analogous post-processing claims for non-DP stability require the relevant stability certificate.

7 Baselines and Comparisons

Exact LASSO selective inference. Lee et al. (2016) gives exact post-selection inference for the LASSO under its Gaussian linear-model and polyhedral-selection conditions. This is a baseline, not a stabilization mechanism, because the selector itself is not randomized or stabilized.

PoSI. Berk et al. (2013) gives simultaneous post-selection coverage over model-dependent targets in the linear-model setting considered there. It is selection-agnostic but can be conservative.

Winner inference without stabilization. Andrews et al. (2024) gives conditional and hybrid confidence procedures for selected winners; Zrnic and Fithian (2024) gives the zoom correction for winner inference. These are baselines for noisy-winner approaches because they correct inference after selection rather than randomizing or stabilizing the winner selector.

Sample splitting. Sample splitting makes inference independent of selection by using separate data. It is broadly valid but sacrifices data efficiency; this motivates stability-based alternatives.

8 Cross-Example Synthesis

8.1 Primitive taxonomy

Primitive	Representative papers	Mechanism and certificate	Utility / suitability
noisy scalar release	Dwork et al. (2006), Zrnic and Jordan (2023)	Laplace/Gaussian noise; DP or stability	RMSE / CI width; atomic primitive
noisy argmax	Zrnic and Jordan (2023), Dwork and Roth (2014), McKenna and Sheldon (2020)	Laplace and permute mechanisms; stability or DP	winner regret; core primitive
noisy top-k	Qiao et al. (2021), Shekelyan and Loukides (2022), Durfee and Rogers (2019)	one-shot noise or thresholding; DP	recall/rank loss; core primitive
noisy threshold	Dwork et al. (2015), Dwork and Roth (2014)	noisy threshold; DP or max-information	pass/fail errors; core primitive
validation comparison for selective inference	Markovic et al. (2017)	Gaussian randomized quality vectors; selective pivots	computation/interval width; strong primitive
DP validation/tuning	Chaudhuri and Vinterbo (2013), Papernot and Steinke (2022)	stable validation or randomized RDP accounting; DP	validation regret/privacy budget; strong primitive

Primitive	Representative papers	Mechanism and certificate	Utility / suitability
support/model selection	Tian and Taylor (2015), Meinshausen and Buehlmann (2010)	randomization or subsampling; selective validity or error control	CI width/PFER; strong primitive
randomized optimization / ERM	Huang et al. (2024), Chaudhuri et al. (2011)	randomized group-regularized objectives for selective likelihood; objective/output perturbation for DP ERM	inference cost or excess risk; useful primitive
subsample aggregation	Meinshausen and Buehlmann (2010), Shah and Samworth (2013), Bach (2008)	subsampling/bootstrapping plus frequency thresholds or support intersection; PFER/low-probability bounds or consistency	PFER/power; strong primitive
private candidate selection	Liu and Talwar (2019)	ϵ_1-DP randomized candidate mechanisms with bounded scores plus selector; DP	candidate utility; strong primitive
reusable holdout	Dwork et al. (2015), Blum and Hardt (2015)	thresholded release; generalization	adaptive accuracy; strong primitive
iterative atom selection	Zrnic and Jordan (2023)	per-step noisy max; composed stability	excess risk; core primitive
best-arm selection	Azize et al. (2024)	private mean estimates and stopping/selection analysis; DP	sample complexity; emerging primitive

8.2 Mechanism taxonomy

Mechanism	Where inserted	Proof / guarantee	Cost / limitation
Laplace noise	scores, thresholds, atoms	density-ratio or sensitivity proof; DP or η-stability	rank error, CI width; global calibration can be pessimistic
Gaussian noise	randomized objectives or statistics	Gaussian selective likelihood or (ϵ, δ)-DP	variance/region width; model-specific inference
Gumbel noise	discrete selection or top-k	exponential-mechanism equivalence; DP	utility regret; not direct PSI by itself
Exponential mechanism	finite candidates	utility sensitivity proof; DP	utility regret; requires a quality function

Mechanism	Where inserted	Proof / guarantee	Cost / limitation
Objective perturbation	ERM objective	density/sensitivity proof; DP	excess risk; optimization assumptions
Output perturbation	ERM output	optimizer sensitivity proof; DP	parameter noise; often needs strong convexity
Subsampling/frequency	feature selectors	selection-frequency bounds; error control	computation/conservativeness; not CI validity
Sparse vector	thresholds	privacy cost controlled by positive-release cap; accuracy depends on threshold margins; DP	pass/fail noise; threshold tuning
Private candidates	model/HPO pipelines	composition over private subroutines; DP	candidate cost; candidates must be private
Regularization	ERM solve	leave-one-out stability; generalization	bias; not usually selection validity

8.3 Proof-strategy taxonomy

The recurring proof strategies are: density-ratio bounds for additive noise; sensitivity calibration for DP; exponential-mechanism utility analysis; coupling for one-shot top-k; sparse-vector accounting for threshold primitives; adaptive composition and post-processing for pipelines; max-information transfer for adaptive inference; selective likelihood conditioning for randomized selective inference; strong convexity for optimizer sensitivity; and selection-frequency bounds for subsampling-based feature selection.

8.4 Implications for a future framework

Clean certificates already exist for noisy argmax, top-k, thresholding, private candidate selection, reusable holdout, and randomized selective-inference selectors, but the certificate types differ: DP/privacy, max-information/generalization, algorithmic stability, and selective likelihood are not interchangeable. DP by itself does not yield confidence intervals or p-values; an inference bridge such as Zrnic–Jordan's stability interval correction, a max-information p-value correction, or a randomized selective likelihood is required. Laplace noise appears in score/threshold/argmax mechanisms; Gaussian noise appears in randomized selective inference and approximate DP; Gumbel/exponential mechanisms appear in finite candidate and top-k selection. The missing abstraction is a modular calculus that maps primitive selectors to stability costs under checkable assumptions and then optimizes noise under downstream inference-width constraints.

Best first theorem. A tractable first theorem would be a heterogeneous or gap-aware noisy argmax/top-k certificate: prove a bound translating candidate-specific sensitivity, gap information, and noise scales into an η-style stability certificate, then plug that certificate into a post-selection

CI correction. This would connect Zrnic–Jordan with private top-k and GAP-MAX-style accuracy analyses, but it is a proposed theorem rather than an established result.

9 TikZ Diagrams

9.1 Noisy argmax

$s_i(D) \to s_i(D) + \xi_i \to \arg\max_i \to \tilde{i} \to$ stability / DP certificate $\to$ inference bridge or utility

9.2 Stability selection

$D \to D^{(1)}, \dots, D^{(B)} \to \widehat{S}^{(1)}, \dots, \widehat{S}^{(B)} \to \hat{\pi}_j \to \{j : \hat{\pi}_j \geq \pi_{\text{thr}}\}$

9.3 Iterative stable selection

$D \to P_1 \to P_2 \to \cdots \to P_T \to \eta_{\text{total}} \leq \sum_t \eta_t$

9.4 Private candidate selection

$D \to$ private candidate generators $\to$ private/stable candidate scores $\to$ private selector $\to \hat{h}$

10 Conclusion

The closest direct match to a future composable framework is Zrnic and Jordan (2023): it already maps noisy selection to a stability certificate and then to post-selection interval correction. Private top-k, Report Noisy Max, the exponential mechanism, and sparse-vector methods provide mature reusable selection primitives. Randomized selective inference provides direct inference but is usually algorithm-specific. Stability selection provides robust support selection but not confidence intervals. The main research gap is a modular calculus that can certify broad classes of heterogeneous pipelines under checkable assumptions, then choose perturbations to optimize downstream inference/utility tradeoffs.

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